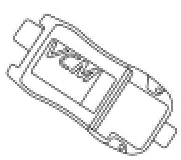
Rear View Mirrors — Interior

Ford Ecat Electronic Parts Catalog

Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

Interior Auto-Dimming Rear View Mirror

The interior rear view mirror with the auto-dimming feature automatically reduces the glare caused by headlamps reflecting in the interior rear view mirror. The auto-dimming feature is disabled when the vehicle is in REVERSE. Power is supplied to the interior auto-dimming mirror when the ignition is in the RUN or ACC position.

The interior auto-dimming rear view mirror has 2 photoelectric sensors that detect forward and rearward light conditions, and based on those inputs, adjusts the reflectance level of the interior rear view mirror and LH exterior mirror to eliminate unwanted glare. The reflectance level of the mirror glass is variable and depends on the amount of rear glare in relation to ambient light conditions in front of the interior mirror.

When the forward sensor detects daytime conditions, the rearward sensor is inactive, and the mirror glass stays in a high reflectance mode. When the forward sensor detects nighttime conditions, the rearward sensor is active and detects glare from the headlights of vehicles approaching from the rear, or other glare producing light sources. To provide increased visibility when backing up, the interior rear view mirror glass will automatically return to a high reflectance mode whenever the gear selector is in REVERSE.

If the forward or rearward sensors are blocked, the auto-dimming interior rear view mirror and LH exterior mirror might not work correctly.

Interior Rear View Mirror With Video Display

For vehicles that are equipped with a rear-mounted object detection camera, a video image will be displayed on the LH side of the interior rear view mirror glass when the gear selector is in REVERSE. For information on the rear-mounted object detection system, refer to Section 413-13.

Inspection and Verification

1. Verify the customer concern.

2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Interior rear view mirror	• Body Control Module (BCM) fuse 32 (15A) • Wiring, terminals or connectors • LH exterior mirror glass • LH exterior mirror • Interior rear view mirror

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. If the concern is not visually evident, verify the symptom and GO to Symptom Chart — Interior Mirror or GO to Symptom Chart — NVH.

Symptom Chart — Interior Mirror

Symptom Chart — Interior Mirror

Condition	Possible Causes	Action
• The interior mirror is blemished	• Interior mirror glass/housing is dirty	• **NOTE:** *Do not clean the housing or glass of any mirror with harsh abrasives, fuel or other petroleum-based cleaning products.* • CLEAN the affected interior mirror surface.
	• The interior mirror is damaged	• INSPECT the interior mirror damage. If needed, INSTALL a new interior rear view mirror. REFER to Interior Rear View Mirror in this section.
• The auto-dimming mirror does not operate correctly	• Obstructed rearward-facing sensor • stickers, window decals, tags • non-OEM window tinting • Obstructed forward-facing sensor • stickers, window decals, tags • non-OEM window tinting	• If possible, REMOVE the obstruction. If the obstruction cannot be removed, GO to Pinpoint Test H to test the interior rear view mirror for correct function.
	• Light source near or inside of vehicle	• None. Any light source the rearward-facing sensor is exposed to can be considered glare.

	• Vehicle inside garage or tunnel • Fuse • Wiring, terminals or connectors • Interior rear view mirror	• None. Ambient light conditions are similar to nighttime. • GO to Pinpoint Test H.
• Interior mirror compass concerns	• —	• GO to Section 413-01.
• Interior mirror video display concerns	• —	• GO to Section 413-13.

Symptom Chart — NVH

Symptom Chart — NVH

Condition	Possible Causes	Action
NOTE: *NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.*		
• The interior mirror vibrates/loose	• Interior mirror mounting loose	• If the mirror is still on the windshield **do not remove.** ATTEMPT to fully seat the mirror first. If the mirror is still loose or vibrates, REMOVE and REINSTALL the mirror. REFER to Interior Rear View Mirror in this section. If the condition still exists, INSTALL a new mirror.

Pinpoint Tests

Pinpoint Test H: The Auto-Dimming Mirror Does Not Operate Correctly

Refer to Wiring Diagrams Cell 124 , Power Mirrors for schematic and connector information.

Normal Operation

The interior auto-dimming mirror contains 2 photoelectric sensors: one in the front of the interior rear view mirror and one mounted on the glass side of the mirror. If the sensors are blocked, the LH exterior mirror and interior mirror auto-dimming feature may not operate correctly. Always verify both sensors are not physically blocked before attempting to diagnose auto-dimming mirror concerns.

For vehicles equipped with an interior mirror with compass display, whenever the gear selector lever is placed in REVERSE the Body Control Module (BCM) communicates selector position to the interior rear view mirror via the Local Interconnect Network (LIN) to disable the auto-dimming feature.

For vehicles equipped with an interior mirror without compass display, whenever the gear selector lever is placed in REVERSE the interior rear view mirror temporarily disables the auto-dimming feature via the reverse circuit input received from the BCM .

The auto-dimming feature is enabled again when the gear selector lever is moved out of REVERSE.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Interior auto-dimming mirror
- LH auto-dimming exterior mirror glass
- LH auto-dimming exterior mirror

PINPOINT TEST H : THE AUTO-DIMMING MIRROR DOES NOT OPERATE CORRECTLY

H1 VERIFY THE FORWARD AND REARWARD FACING SENSORS ARE NOT BLOCKED

- Visually verify the forward and rearward facing sensors are not blocked. Sources of blockage can include:
  - stickers, window decals or tags.
  - fold-down screens for TVs or DVD players.
  - non-OEM window tinting.

Were either of the sensors blocked?

Yes	If possible, REMOVE the blockage. TEST the system for normal operation. If it is not possible to remove the blockage, ADVISE the customer of the correct operation of the interior auto-dimming mirror.
No	GO to H2.

H2 CHECK OPERATION OF THE BACKUP LAMPS

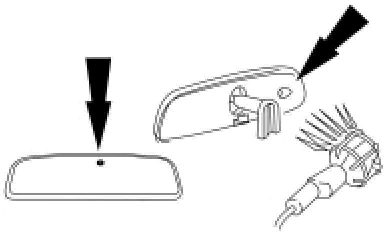
- Ignition ON.
- Move the selector lever through the entire range.

Do the backup lamps illuminate and only in REVERSE?

Yes	If both the LH exterior mirror and interior mirror do not dim correctly, GO to H3. If only the LH exterior mirror does not dim correctly, GO to H9. If only the interior mirror does not dim correctly: — For vehicles equipped with an interior mirror with compass display, GO to H17. — For vehicles equipped an interior mirror without compass display, GO to H18.
No	REFER to Section 417-01 to diagnose the backup lamps.

H3 CHECK OPERATION OF THE INTERIOR AUTO-DIMMING MIRROR — DAYLIGHT CONDITIONS

- Use a bright lamp to illuminate the forward facing sensor. The mirror should adjust to a high reflectance mode (mirror will be clear).



- N0116951

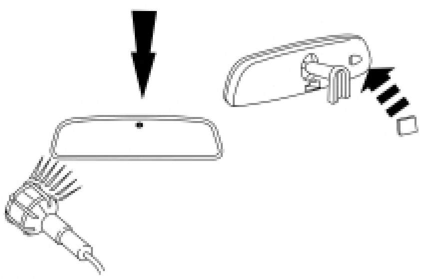
Does the mirror adjust to the high reflectance (clear) mode?

Yes	GO to H4.
No	INSTALL a new interior rear view mirror. REFER to Interior Rear View Mirror in this section. TEST the system for normal operation.

H4 CHECK OPERATION OF THE INTERIOR AUTO-DIMMING MIRROR — NIGHTTIME CONDITIONS WITH GLARE

- **NOTE:** *Covering the sensor with a finger or hand is not adequate.*

  Simulate nighttime conditions with glare:
  - cover the forward sensor with black electrical tape or other dark material.
  - illuminate the rearward facing sensor. The mirror should darken to a lower reflectance mode.



- N0115902

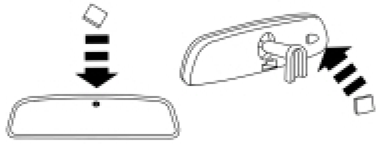
Did the mirror darken to a lower reflectance (darker) mode?

Yes	GO to H5.
No	GO to H7.

H5 CHECK OPERATION OF THE INTERIOR AUTO-DIMMING MIRROR — NIGHTTIME CONDITIONS WITHOUT GLARE

- **NOTE:** *Covering the sensor with a finger or hand is not adequate.*

  Simulate nighttime conditions without glare:
  - cover the rearward facing sensor. The mirror should adjust to the high reflectance mode.



- N0116353

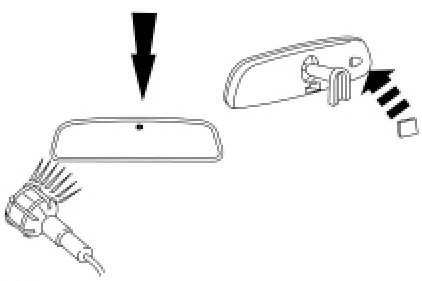
Did the mirror adjust to the high reflectance (clear) mode?

Yes	GO to H6.
No	GO to H7.

H6 CHECK OPERATION OF THE INTERIOR AUTO-DIMMING MIRROR — NIGHTTIME CONDITIONS WITH THE VEHICLE IN REVERSE

- **NOTE:** *Covering the sensor with a finger or hand is not adequate*.

  Simulate nighttime conditions with glare:
  - cover the forward sensor with black electrical tape or other dark material.
  - illuminate the rearward facing sensor. (mirror should darken at this time).



- N0115902

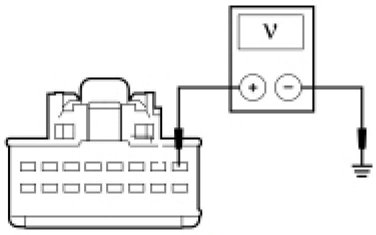
- Select REVERSE.

Did the mirror adjust to a high reflectance (clear) mode?

Yes	The system is operating normally at this time. ADVISE the customer of the correct operation of the interior auto-dimming mirror.
No	GO to H7.

H7 CHECK FOR VOLTAGE TO THE INTERIOR AUTO-DIMMING MIRROR

- Ignition OFF.
- Disconnect: Interior Auto-Dimming Mirror C9039.
- Ignition ON.
- Measure the voltage between interior auto-dimming mirror C9039 Pin 1, circuit CBP32 (GN/VT), harness side and ground.

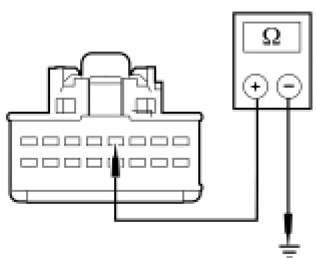
- 

N0121931

Is the voltage greater than 11 volts?

Yes	GO to H8.
No	VERIFY that Body Control Module (BCM) fuse 32 (15A) is OK. If OK, REPAIR the circuit for an open. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

H8 CHECK FOR GROUND TO THE INTERIOR AUTO-DIMMING MIRROR

- Ignition OFF.
- Measure the resistance between interior auto-dimming mirror C9039 Pin 4, circuit GD133 (BK), harness side and ground.
- 

N0121933

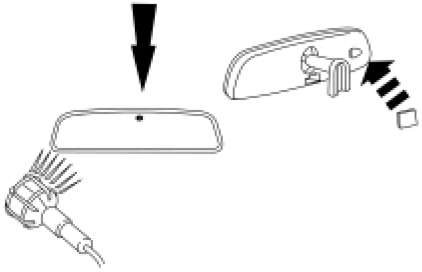
Is the resistance less than 3 ohms?

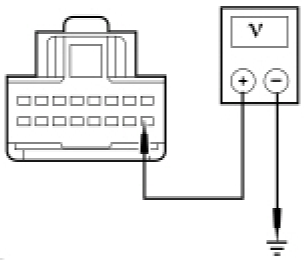
Yes	INSTALL new interior rear view mirror. REFER to Interior Rear View Mirror in this section.
No	REPAIR the circuit. TEST the system for normal operation.

H9 CHECK THE INTERIOR REAR VIEW MIRROR OUTPUT TO THE LH EXTERIOR REAR VIEW MIRROR

- Ignition OFF.
- Disconnect: LH Exterior Rear View Mirror C521.
- Ignition ON.
- **NOTE:** *Covering the sensor with a finger or hand is not adequate.*

  Simulate nighttime conditions with glare:
  - cover the forward facing sensor with black electrical tape or other dark material.
  - illuminate the rearward facing sensor. The interior mirror should darken to a lower reflectance mode.

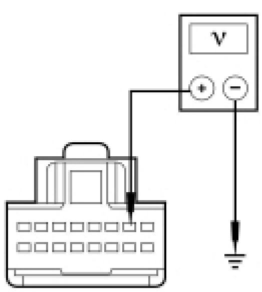
- 

- Measure the voltage between LH exterior mirror C521 Pin 9, circuit LRD12 (BU/GY), harness side and ground.
- 

Is the voltage greater than 1 volt?

Yes	GO to H10.
No	GO to H15.

H10 CHECK THE GROUND CIRCUIT BETWEEN THE LH EXTERIOR REAR VIEW MIRROR AND INTERIOR REAR VIEW MIRROR FOR A SHORT TO VOLTAGE

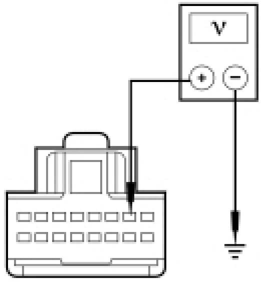
- Measure the voltage between LH exterior rear view mirror C521 Pin 2, circuit RRD12 (BN), harness side and ground.
- 

Is any voltage present?

Yes	GO to H11.
No	GO to H12.

H11 ISOLATE THE SHORT TO VOLTAGE ON THE GROUND CIRCUIT

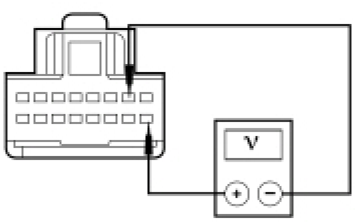
- Ignition OFF.

- Disconnect: Interior Rear View Mirror C9039.
- Ignition ON.
- Measure the voltage between LH exterior mirror C521 Pin 2, circuit RRD12 (BN), harness side and ground.
- 
N0124670

Is any voltage present?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	INSTALL a new interior rear view mirror. REFER toInterior Rear View Mirror in this section. TEST the system for normal operation.

H12 CHECK THE GROUND CIRCUIT FOR AN OPEN

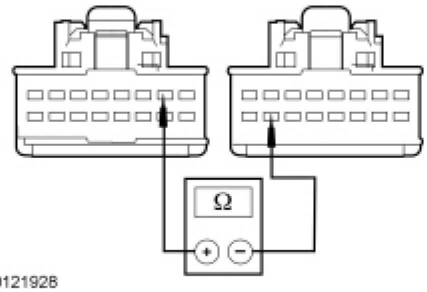
- Measure the voltage between LH exterior mirror C521 Pin 9, circuit LRD12 (BN), harness side and LH exterior mirror C521 Pin 2, circuit RRD12 (BN), harness side.
- 
N0124671

Is the voltage greater than 1 volt?

Yes	CHECK the LH exterior mirror jumper harness between the vehicle harness and the exterior mirror glass for shorted or open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new LH exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a new LH exterior mirror glass. REFER to Exterior Mirror Glass — Trailer Tow in this section. TEST the system for normal operation.
No	GO to H13.

H13 CHECK THE GROUND CIRCUIT BETWEEN THE EXTERIOR REAR VIEW MIRROR AND INTERIOR AUTO-DIMMING MIRROR FOR AN OPEN

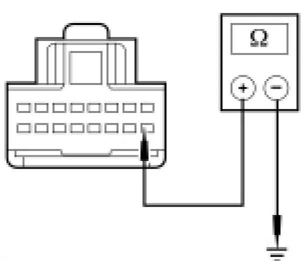
- Ignition OFF.

- Disconnect: Interior Rear View Mirror C9039.
- Measure the resistance between interior rear view mirror C9039 Pin 15, circuit RRD12 (BN), harness side and LH exterior mirror C521 Pin 2, circuit RRD12 (BN), harness side.
- 

Is the resistance less than 3 ohms?

Yes	GO to H14.
No	REPAIR the circuit for an open. TEST the system for normal operation.

H14 CHECK THE CIRCUIT BETWEEN THE LH EXTERIOR REAR VIEW MIRROR AND INTERIOR REAR VIEW MIRROR FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: Interior Rear View Mirror C9039.
- Measure the resistance between LH exterior mirror C521 Pin 9, circuit LRD12 (BU/GY), harness side and ground.
- 

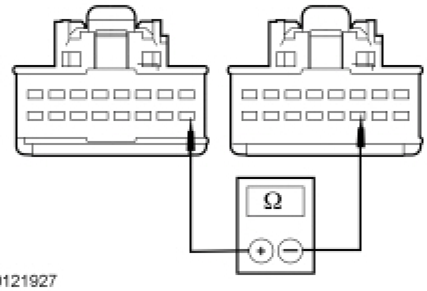
Is the resistance greater than 10,000 ohms?

Yes	GO to H15.
No	REPAIR the circuit. TEST the system for normal operation.

H15 CHECK THE CIRCUIT BETWEEN THE LH EXTERIOR REAR VIEW MIRROR AND INTERIOR REAR VIEW MIRROR FOR AN OPEN

- Ignition OFF.
- Disconnect: Interior Rear View Mirror C9039 (if not previously disconnected) .

- Measure the resistance between interior rear view mirror C9039 Pin 11, circuit LRD12 (BU/GY), harness side and LH exterior mirror C521 Pin 9, circuit LRD12 (BU/GY), harness side.



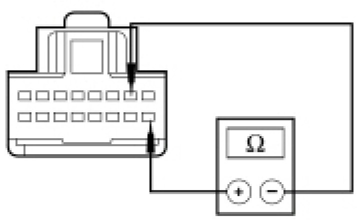
- N0121927

Is the resistance less than 3 ohms?

Yes	GO to H16.
No	REPAIR the circuit for an open. TEST the system for normal operation.

H16 CHECK FOR A SHORT BETWEEN THE LH EXTERIOR REAR VIEW MIRROR AND INTERIOR REAR VIEW MIRROR CIRCUITS

- Measure the resistance between LH rear view mirror C521 Pin 2, circuit RRD12 (BU/GY), and LH exterior mirror C521 Pin 9, circuit LRD12 (BU/GY), harness side.



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Is the resistance greater than 10,000 ohms?

Yes	INSTALL a new interior rear view mirror. REFER to Interior Rear View Mirror in this section. TEST the system for normal operation.
No	REPAIR the circuits. TEST the system for normal operation.

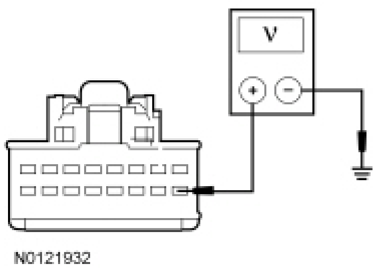
H17 CHECK THE LIN CIRCUIT FOR CORRECT OPERATION

- Position the vehicle and observe the compass display.

Does the compass display correctly?

Yes	INSTALL a new interior rear view mirror. REFER to Interior Rear View Mirror in this section. TEST the system for normal operation.
No	REFER to Section 413-01 to diagnose the Local Interconnect Network (LIN) concern.

H18 CHECK FOR VOLTAGE TO THE INTERIOR AUTO-DIMMING MIRROR REVERSE CIRCUIT

- Ignition OFF.
- Disconnect: Interior Rear View Mirror C9039.
- Ignition ON.
- Select REVERSE.
- Measure the voltage between interior rear view mirror C9039 Pin 9, circuit CLS28 (BU/WH), harness side and ground.
- 

Is the voltage greater than 11 volts?

Yes	INSTALL a new interior rear view mirror. REFER to Interior Rear View Mirror in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.